DS 4420 Algorithmic Mountain Bike Trail Difficulty Classification

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Introduction

Rating mountain bike trails with objective difficulty ratings is an open question largely left unanswered. This project aims to explain multiple machine learning models built to classify the difficulty of existing mountain bike trail data as well as evaluate those models. The models showcased fall broadly in the categories of Multi-layer Perceptrons, Convolutional Neural Networks, and Bayesian Machine Learning.

Motivation and Data

Why MLPs, CNNs, and Bayesian ML Models

The dataset used in this project is comprised of mountain bike trail data for more than 30,000 trails scraped from the **MTB Project website**. The dataset contains GPS (latitude, longitude, and elevation), elevation loss, elevation gain, max grade, average grade, and more features. MLPs are relatively simple machine learning models, but they can't learn complex data relationships very well. GPS data can be visualized relatively easily and convolutional neural networks (CNN) mostly use image data allowing for a human-interactable model. Finally, experimenting with multi-class Bayesian machine learning models would allow for further discovery of feature impact on output predictions [1].

On the right is a bike trail GPS image. GPS data has been run through the image converter and the elevation heatmap has been scaled to that of the image. This image has 1 color channel and is 256x256 pixels.

Fig. 1: Bike trail 7026157 GPS image

Bike trail difficulties in this data set are skewed towards intermediate ratings. A neural network trained on the raw splits of difficulties will likely learn to guess the majority class everytime. To avoid this issue the data set was balanced. Another issue with creating image data from GPS data was whether to do global or local elevation heatmapping for the trail images. After testing CNN performance on elevations scaled on the global dataset and per-image, the latter was chosen for its efficacy.

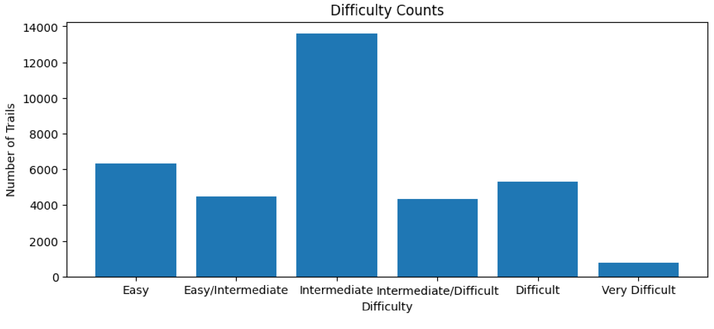

Fig. 2: Difficulty distributions from trail dataset

Methodology

MLP and CNN models each have 3 dense layers before a final softmax output layer. The CNN has 2 convolutional layers with 3x3 kernels with stride of 1 followed by 5x5 max pooling layers. The CNN is provided 4 additional features of elevation gain, elevation loss, max grade, and average grade after flattening to improve performance.

Fig. 3: Trail image

Fig. 4: Trail image passed through second convolutional layer's kernel

Bayesian Model is constructed with the *Difficult* class as the reference class.

Likelihood Class probabilities are given by the softmax over weight matrix W:

$$\log p(\mathbf{y} \mid X, W) = \sum_{i=1}^{n} \Big[\eta_{i,y_i} - \log \sum_{k=1}^{K} \exp(\eta_{i,k}) \Big] \tag{1}$$

Prior Isotropic Gaussian distributions on each weight with $\sigma_0 = 3$:

$$p(W) = \prod_{p,k} \mathcal{N}(w_{pk} \mid 0,\, \sigma_0^2 I) \tag{2}$$

Posterior:

$$p(W \mid \mathbf{y}, X) \;\propto\; p(\mathbf{y} \mid X, W)\, p(W)$$

Acceptance ratio for Metropolis sampling

$$\alpha = \exp\big[\log p(W' \mid \mathbf{y}, X) - \log p(W \mid \mathbf{y}, X)\big] \tag{3}$$

Model ran $1,500,000$ iterations with 18% burn-in and thinning every 200th sample.

Results

MLP

The MLP has a test accuracy similar to the those of the CNN and Bayesian model. This stands to substantiate claims made in Langer et al. [1] implying that without specialized IMU data to backup the CNN, it won't outperform more basic models.

CNN

The convolutional layer kernels produce images that are rather sparse, but without those layers, results are drastically worse. The model has extracted what signal there is for predicting classes but there's still noise in the data.

Bayesian Model

The strongest feature for making predictions is the average grade of the trail, with *Easy* difficulty prior distributions having a -1.684 averaged weight value over this feature.

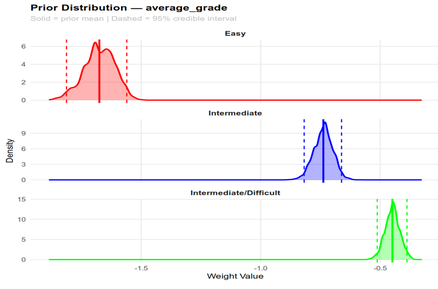

Fig. 5: Prior weight distribution for `average_grade` feature

Results

Model	Training Accuracy	Test Accuracy
MLP	0.534	0.500
CNN	0.464	0.448
Bayesian	0.435	0.445

Table 1: Model Performance Comparison

Discussion and Web App

The model's performances fell within expectations. Accuracies of around 44%-50% across all three models imply that the models are doing around 20% better than just random guessing (on average). It's interesting to see that convolutions improved accuracy as they are rather sparse. Confusion matrices for each model was generated and, in particular, the Bayesian model seems to be good at classifying *Easy* and *Difficult* bike trails but has difficulty with differentiating the other classes. This is most likely because *Difficult* is the reference class and *Easy* should be the furthest class from the reference.

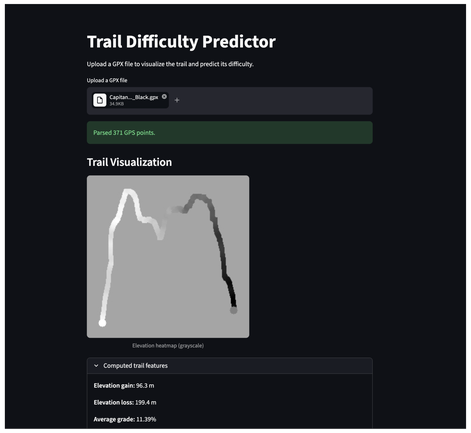

Fig. 6: MTB-Classify Landing Page

Fig. 7: QR to MTB-Classify

Above is an image of an interactive webapp to try out the CNN model. Uploading '`.gpx`' files shows the trail as the greyscale, locally heatmapped image. Below the image, and the parsed features, you can see the prediction percentages for the four classes, and what the model ultimately predicts the route difficulty as.

References

[1] S. Langer et al. "Difficulty classification of mountainbike downhill trails utilizing Deep Neural Networks". In: *Communications in Computer and Information Science* 1168 (Mar. 2020), pp. 270–280. DOI: 10.1007/978-3-030-43887-6_21.